

```
C:\WINDOWS\system32\cmd. x + v

C:\Users\Sanjana CA\OneDrive\Documents\CynarisInternship\OmniRetail_AI>python -m pytest tests -v
===== test session starts =====
platform win32 -- Python 3.14.0, pytest-9.1.1, pluggy-1.6.0 -- C:\Users\Sanjana CA\AppData\Local\Programs\Python\Python314\python.exe
cachedir: .pytest_cache
rootdir: C:\Users\Sanjana CA\OneDrive\Documents\CynarisInternship\OmniRetail_AI
plugins: anyio-4.14.2, langsmith-0.10.17, asyncio-1.4.0
asyncio: mode=Mode.STRICT, debug=False, asyncio_default_fixture_loop_scope=None, asyncio_default_test_loop_scope=function
collected 5 items

tests/test_data_validation.py::test_reviews_dataset_exists_and_is_valid PASSED [ 20%]
tests/test_data_validation.py::test_reviews_dataset_has_no_missing_values PASSED [ 40%]
tests/test_data_validation.py::test_ecommerce_test_cases_exist_and_are_valid PASSED [ 60%]
tests/test_data_validation.py::test_data_loader_reviews PASSED [ 80%]
tests/test_data_validation.py::test_data_loader_ecommerce_tests PASSED [100%]

===== 5 passed in 1.99s =====

C:\Users\Sanjana CA\OneDrive\Documents\CynarisInternship\OmniRetail_AI>python app\data_validation.py
=== OmniRetail AI Data Validation ===

Product Reviews Dataset
Shape: (800, 11)
Columns: ['review_id', 'product_id', 'category', 'rating', 'review_date', 'verified_purchase', 'helpful_votes', 'review_length_chars',
'contains_image', 'sentiment', 'review_text']
Missing values: 0

E-Commerce Test Cases
Records: 50
Fields: ['test_id', 'input', 'expected_label', 'model_target']

Clickstream Dataset: Pending

Data validation completed successfully.

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Fields: ['test_id', 'input', 'expected_label', 'model_target']

Clickstream Dataset: Pending

Data validation completed successfully.

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collected 5 items

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tests/test_data_validation.py::test_reviews_dataset_has_no_missing_values PASSED [ 40%]
tests/test_data_validation.py::test_ecommerce_test_cases_exist_and_are_valid PASSED [ 60%]
tests/test_data_validation.py::test_data_loader_reviews PASSED [ 80%]
tests/test_data_validation.py::test_data_loader_ecommerce_tests PASSED [100%]

===== 5 passed in 2.16s =====

C:\Users\Sanjana CA\OneDrive\Documents\CynarisInternship\OmniRetail_AI>python -c "from app.data_loader import load_product_reviews, load_ecommerce_tests; print('Reviews:', load_product_reviews().shape); print('Test cases:', len(load_ecommerce_tests()))"
Reviews: (800, 11)
Test cases: 50

C:\Users\Sanjana CA\OneDrive\Documents\CynarisInternship\OmniRetail_AI>C:\Users\Sanjana CA\OneDrive\Documents\CynarisInternship\OmniRetail_AI>
The syntax of the command is incorrect.

C:\Users\Sanjana CA\OneDrive\Documents\CynarisInternship\OmniRetail_AI>C:\Users\Sanjana CA\OneDrive\Documents\CynarisInternship\OmniRetail_AI>
The syntax of the command is incorrect.

C:\Users\Sanjana CA\OneDrive\Documents\CynarisInternship\OmniRetail_AI>
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plugins: anyio-4.14.2, langsmith-0.10.17, asyncio-1.4.0
asyncio: mode=Mode.STRICT, debug=False, asyncio_default_fixture_loop_scope=None, asyncio_default_test_loop_scope=function
collected 3 items

tests/test_data_validation.py::test_reviews_dataset_exists_and_is_valid PASSED [ 33%]
tests/test_data_validation.py::test_reviews_dataset_has_no_missing_values PASSED [ 66%]
tests/test_data_validation.py::test_ecommerce_test_cases_exist_and_are_valid PASSED [100%]

===== 3 passed in 1.61s =====

C:\Users\Sanjana CA\OneDrive\Documents\CynarisInternship\OmniRetail_AI>dir app
Volume in drive C is Windows-SSD
Volume Serial Number is EA8D-9B22

Directory of C:\Users\Sanjana CA\OneDrive\Documents\CynarisInternship\OmniRetail_AI\app

04-10-2026 19:19 <DIR>      .
04-10-2026 19:21 <DIR>      ..
04-10-2026 19:12          0 config.py
04-10-2026 19:12          0 data_loader.py
04-10-2026 19:19      1,387 data_validation.py
04-10-2026 19:12          0 __init__.py
                4 File(s)      1,387 bytes
                2 Dir(s)  11,995,820,032 bytes free

C:\Users\Sanjana CA\OneDrive\Documents\CynarisInternship\OmniRetail_AI>dir docs
Volume in drive C is Windows-SSD
Volume Serial Number is EA8D-9B22

Directory of C:\Users\Sanjana CA\OneDrive\Documents\CynarisInternship\OmniRetail_AI\docs
```

```
C:\WINDOWS\system32\cmd. x + v

Directory of C:\Users\Sanjana CA\OneDrive\Documents\CynarisInternship\OmniRetail_AI\docs

04-10-2026 19:12 <DIR>      .
04-10-2026 19:21 <DIR>      ..
04-10-2026 19:16      11,013 ARCHITECTURE.md
04-10-2026 19:15      5,134 DATA_DICTIONARY.md
04-10-2026 19:17      8,645 RESEARCH.md
                3 File(s)      24,792 bytes
                2 Dir(s)  11,992,485,888 bytes free

C:\Users\Sanjana CA\OneDrive\Documents\CynarisInternship\OmniRetail_AI>dir tests
Volume in drive C is Windows-SSD
Volume Serial Number is EA8D-9B22

Directory of C:\Users\Sanjana CA\OneDrive\Documents\CynarisInternship\OmniRetail_AI\tests

04-10-2026 19:21 <DIR>      .
04-10-2026 19:21 <DIR>      ..
04-10-2026 19:21      1,120 test_data_validation.py
04-10-2026 19:12          0 __init__.py
04-10-2026 19:21 <DIR>      __pycache__
                2 File(s)      1,120 bytes
                3 Dir(s)  11,985,268,736 bytes free

C:\Users\Sanjana CA\OneDrive\Documents\CynarisInternship\OmniRetail_AI>dir data
Volume in drive C is Windows-SSD
Volume Serial Number is EA8D-9B22

Directory of C:\Users\Sanjana CA\OneDrive\Documents\CynarisInternship\OmniRetail_AI\data

04-10-2026 19:09 <DIR>      .
04-10-2026 19:21 <DIR>      ..
04-10-2026 18:53      12,091 aimpl_ecommerce_tests.json
04-10-2026 18:53      88,574 aimpl_product_reviews.csv
                2 File(s)      100,665 bytes
                2 Dir(s)  11,980,640,256 bytes free
```

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C:\WINDOWS\system32\cmd. x + v
04-10-2026 19:21 <DIR>      _pycache__
                2 File(s)      1,120 bytes
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04-10-2026 19:09 <DIR>      .
04-10-2026 19:21 <DIR>      ..
04-10-2026 18:53                12,091 aml_ecommerce_tests.json
04-10-2026 18:53                88,574 aml_product_reviews.csv
                2 File(s)      100,665 bytes
                2 Dir(s)  11,980,640,256 bytes free

C:\Users\Sanjana CA\OneDrive\Documents\CynarisInternship\OmniRetail_AI>notepad app\config.py
C:\Users\Sanjana CA\OneDrive\Documents\CynarisInternship\OmniRetail_AI>notepad app\data_loader.py
C:\Users\Sanjana CA\OneDrive\Documents\CynarisInternship\OmniRetail_AI>python -c "from app.data_loader import load_product_reviews, load_ecommerce_tests; pr
int('Reviews:', load_product_reviews().shape); print('Test cases:', len(load_ecommerce_tests()))"
Reviews: (880, 11)
Test cases: 50

C:\Users\Sanjana CA\OneDrive\Documents\CynarisInternship\OmniRetail_AI>notepad tests\test_data_validation.py
C:\Users\Sanjana CA\OneDrive\Documents\CynarisInternship\OmniRetail_AI>python -m pytest tests -v
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